MONASH University

Semester One 2023
Examination Period

Faculty of Business & Economics

UNIT CODES:	**ETF3231/ETF5231**
TITLE OF PAPER:	**Business Forecasting**
EXAM DURATION:	2 hours 10 minutes

AUTHORISED MATERIALS

This is a closed book exam, with the following permitted items.

- A physical calculator of any type or Virtual Calculator:
  - Inbuilt Mac/Windows calculator
  - Website https://www.educalc.net/2336211.page
  - 10bii Financial Calculator for Mac by K2 Cashflow, https://apps.apple.com/au/app/10bii-financial-calculator/id473144920
- 5 blank pages for use as working sheets
- 2 pre-printed answer sheets

RULES

During your eExam, you must not have in your possession any item/material that has not been authorised for your exam. This includes books, notes, paper, electronic device/s, smart watch/device, or writing on any part of your body. Authorised items are listed above. Items/materials on your device, desk, chair, in your clothing or otherwise on your person will be deemed to be in your possession. Mobile phones must be switched off and placed face-down on your desk during your exam attempt.

You must not retain, copy, memorise or note down any exam content for personal use or to share with any other person by any means during or following your exam. You are not allowed to copy/paste text to or from external sources unless this has been authorised by your Chief Examiner.

You must comply with any instructions given to you by Monash exam staff.

As a student, and under Monash University's Student Academic Integrity procedure, you must undertake all your assessments with honesty and integrity. You must not allow anyone else to do work for you and you must not do any work for others. You must not contact, or attempt to contact, another person in an attempt to gain unfair advantage during your assessment. Assessors may take reasonable steps to check that your work displays the expected standards of academic integrity.

Failure to comply with the above instructions, or attempting to cheat or cheating in an assessment may constitute a breach of instructions under regulation 23 of the Monash University (Academic Board) Regulations or may constitute an act of academic misconduct under Part 7 of the Monash University (Council) Regulations.

The exam contains FIVE questions. ALL questions must be answered. The exam is worth 100 marks in total.

Below are the State Space equations for each of the models in the ETS framework.

ADDITIVE ERROR MODELS

Trend	Seasonal: N	Seasonal: A	Seasonal: M
N	$y_t = \ell_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha\varepsilon_t$	$y_t = \ell_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha\varepsilon_t$ $s_t = s_{t-m} + \gamma\varepsilon_t$	$y_t = \ell_{t-1}s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \alpha\varepsilon_t/s_{t-m}$ $s_t = s_{t-m} + \gamma\varepsilon_t/\ell_{t-1}$
A	$y_t = \ell_{t-1} + b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha\varepsilon_t$ $b_t = b_{t-1} + \beta\varepsilon_t$	$y_t = \ell_{t-1} + b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha\varepsilon_t$ $b_t = b_{t-1} + \beta\varepsilon_t$ $s_t = s_{t-m} + \gamma\varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1})s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha\varepsilon_t/s_{t-m}$ $b_t = b_{t-1} + \beta\varepsilon_t/s_{t-m}$ $s_t = s_{t-m} + \gamma\varepsilon_t/(\ell_{t-1} + b_{t-1})$
A_d	$y_t = \ell_{t-1} + \phi b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha\varepsilon_t$ $b_t = \phi b_{t-1} + \beta\varepsilon_t$	$y_t = \ell_{t-1} + \phi b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha\varepsilon_t$ $b_t = \phi b_{t-1} + \beta\varepsilon_t$ $s_t = s_{t-m} + \gamma\varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1})s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha\varepsilon_t/s_{t-m}$ $b_t = \phi b_{t-1} + \beta\varepsilon_t/s_{t-m}$ $s_t = s_{t-m} + \gamma\varepsilon_t/(\ell_{t-1} + \phi b_{t-1})$
M	$y_t = \ell_{t-1}b_{t-1} + \varepsilon_t$ $\ell_t = \ell_{t-1}b_{t-1} + \alpha\varepsilon_t$ $b_t = b_{t-1} + \beta\varepsilon_t/\ell_{t-1}$	$y_t = \ell_{t-1}b_{t-1} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1}b_{t-1} + \alpha\varepsilon_t$ $b_t = b_{t-1} + \beta\varepsilon_t/\ell_{t-1}$ $s_t = s_{t-m} + \gamma\varepsilon_t$	$y_t = \ell_{t-1}b_{t-1}s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1}b_{t-1} + \alpha\varepsilon_t/s_{t-m}$ $b_t = b_{t-1} + \beta\varepsilon_t/(s_{t-m}\ell_{t-1})$ $s_t = s_{t-m} + \gamma\varepsilon_t/(\ell_{t-1}b_{t-1})$
M_d	$y_t = \ell_{t-1}b_{t-1}^{\phi} + \varepsilon_t$ $\ell_t = \ell_{t-1}b_{t-1}^{\phi} + \alpha\varepsilon_t$ $b_t = b_{t-1}^{\phi} + \beta\varepsilon_t/\ell_{t-1}$	$y_t = \ell_{t-1}b_{t-1}^{\phi} + s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1}b_{t-1}^{\phi} + \alpha\varepsilon_t$ $b_t = b_{t-1}^{\phi} + \beta\varepsilon_t/\ell_{t-1}$ $s_t = s_{t-m} + \gamma\varepsilon_t$	$y_t = \ell_{t-1}b_{t-1}^{\phi}s_{t-m} + \varepsilon_t$ $\ell_t = \ell_{t-1}b_{t-1}^{\phi} + \alpha\varepsilon_t/s_{t-m}$ $b_t = b_{t-1}^{\phi} + \beta\varepsilon_t/(s_{t-m}\ell_{t-1})$ $s_t = s_{t-m} + \gamma\varepsilon_t/(\ell_{t-1}b_{t-1}^{\phi})$

MULTIPLICATIVE ERROR MODELS

Trend	Seasonal: N	Seasonal: A	Seasonal: M
N	$y_t = \ell_{t-1}(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}(1 + \alpha\varepsilon_t)$	$y_t = (\ell_{t-1} + s_{t-m})(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \alpha(\ell_{t-1} + s_{t-m})\varepsilon_t$ $s_t = s_{t-m} + \gamma(\ell_{t-1} + s_{t-m})\varepsilon_t$	$y_t = \ell_{t-1}s_{t-m}(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}(1 + \alpha\varepsilon_t)$ $s_t = s_{t-m}(1 + \gamma\varepsilon_t)$
A	$y_t = (\ell_{t-1} + b_{t-1})(1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1})(1 + \alpha\varepsilon_t)$ $b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1})\varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1} + s_{t-m})(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + b_{t-1} + \alpha(\ell_{t-1} + b_{t-1} + s_{t-m})\varepsilon_t$ $b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1} + s_{t-m})\varepsilon_t$ $s_t = s_{t-m} + \gamma(\ell_{t-1} + b_{t-1} + s_{t-m})\varepsilon_t$	$y_t = (\ell_{t-1} + b_{t-1})s_{t-m}(1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + b_{t-1})(1 + \alpha\varepsilon_t)$ $b_t = b_{t-1} + \beta(\ell_{t-1} + b_{t-1})\varepsilon_t$ $s_t = s_{t-m}(1 + \gamma\varepsilon_t)$
A_d	$y_t = (\ell_{t-1} + \phi b_{t-1})(1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1})(1 + \alpha\varepsilon_t)$ $b_t = \phi b_{t-1} + \beta(\ell_{t-1} + \phi b_{t-1})\varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1} + s_{t-m})(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1} + \phi b_{t-1} + \alpha(\ell_{t-1} + \phi b_{t-1} + s_{t-m})\varepsilon_t$ $b_t = \phi b_{t-1} + \beta(\ell_{t-1} + \phi b_{t-1} + s_{t-m})\varepsilon_t$ $s_t = s_{t-m} + \gamma(\ell_{t-1} + \phi b_{t-1} + s_{t-m})\varepsilon_t$	$y_t = (\ell_{t-1} + \phi b_{t-1})s_{t-m}(1 + \varepsilon_t)$ $\ell_t = (\ell_{t-1} + \phi b_{t-1})(1 + \alpha\varepsilon_t)$ $b_t = \phi b_{t-1} + \beta(\ell_{t-1} + \phi b_{t-1})\varepsilon_t$ $s_t = s_{t-m}(1 + \gamma\varepsilon_t)$
M	$y_t = \ell_{t-1}b_{t-1}(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}b_{t-1}(1 + \alpha\varepsilon_t)$ $b_t = b_{t-1}(1 + \beta\varepsilon_t)$	$y_t = (\ell_{t-1}b_{t-1} + s_{t-m})(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}b_{t-1} + \alpha(\ell_{t-1}b_{t-1} + s_{t-m})\varepsilon_t$ $b_t = b_{t-1} + \beta(\ell_{t-1}b_{t-1} + s_{t-m})\varepsilon_t/\ell_{t-1}$ $s_t = s_{t-m} + \gamma(\ell_{t-1}b_{t-1} + s_{t-m})\varepsilon_t$	$y_t = \ell_{t-1}b_{t-1}s_{t-m}(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}b_{t-1}(1 + \alpha\varepsilon_t)$ $b_t = b_{t-1}(1 + \beta\varepsilon_t)$ $s_t = s_{t-m}(1 + \gamma\varepsilon_t)$
M_d	$y_t = \ell_{t-1}b_{t-1}^{\phi}(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}b_{t-1}^{\phi}(1 + \alpha\varepsilon_t)$ $b_t = b_{t-1}^{\phi}(1 + \beta\varepsilon_t)$	$y_t = (\ell_{t-1}b_{t-1}^{\phi} + s_{t-m})(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}b_{t-1}^{\phi} + \alpha(\ell_{t-1}b_{t-1}^{\phi} + s_{t-m})\varepsilon_t$ $b_t = b_{t-1}^{\phi} + \beta(\ell_{t-1}b_{t-1}^{\phi} + s_{t-m})\varepsilon_t/\ell_{t-1}$ $s_t = s_{t-m} + \gamma(\ell_{t-1}b_{t-1}^{\phi} + s_{t-m})\varepsilon_t$	$y_t = \ell_{t-1}b_{t-1}^{\phi}s_{t-m}(1 + \varepsilon_t)$ $\ell_t = \ell_{t-1}b_{t-1}^{\phi}(1 + \alpha\varepsilon_t)$ $b_t = b_{t-1}^{\phi}(1 + \beta\varepsilon_t)$ $s_t = s_{t-m}(1 + \gamma\varepsilon_t)$

SECTION A

Write about a quarter of a page each on any **four** of the following topics.

1. Linear regression models are simplistic because the real world is nonlinear.

   5 marks

2. MAE, RMSE, MAPE and MASE are all similar measures; hence it does not matter which one we use.

   5 marks

3. Always choose the model with the smallest sum of squared errors.

   5 marks

4. A good test of whether a model will produce good forecasts is that the residuals are stationary.

   5 marks

5. Forecasts should never give the same value for all forecast horizons.

   5 marks

6. Regression models are not useful for forecasting because we always need to provide forecasts of the predictors.

   5 marks

Total: 20 marks

SECTION B

Figures 1–3 relate to the weekly mortality rate in Australia from 2015 – 2022. The mortality rate is defined as the number of deaths per thousand people in Australia in each week.

```
aus_mortality |>
  autoplot(rate) +
  labs(title = "Weekly Australian mortality rate", y = "Deaths per thousand people")
```

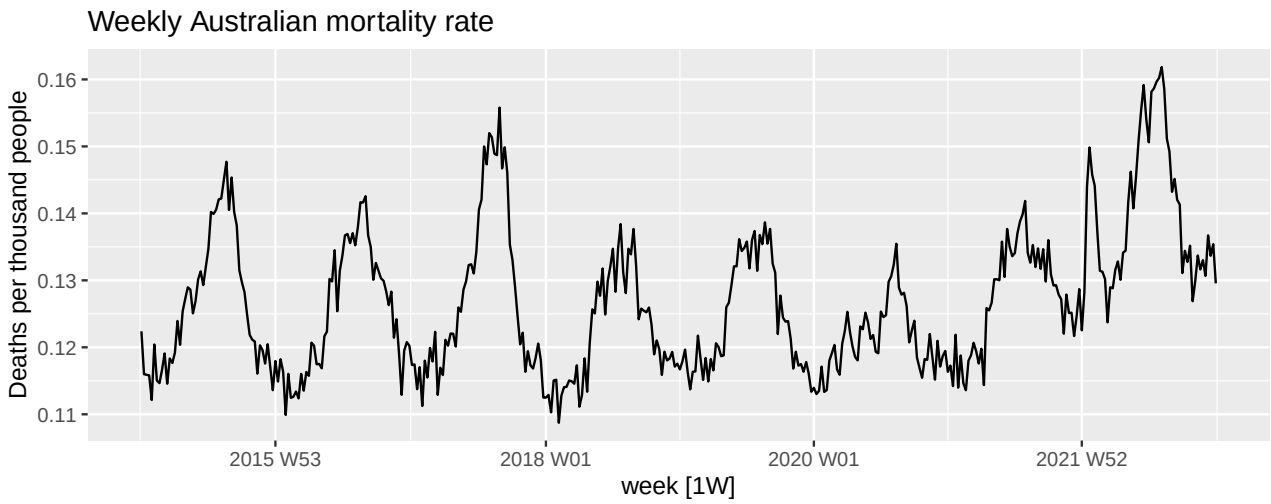

Figure 1

```
aus_mortality |>
  gg_season(rate, labels = "right")
```

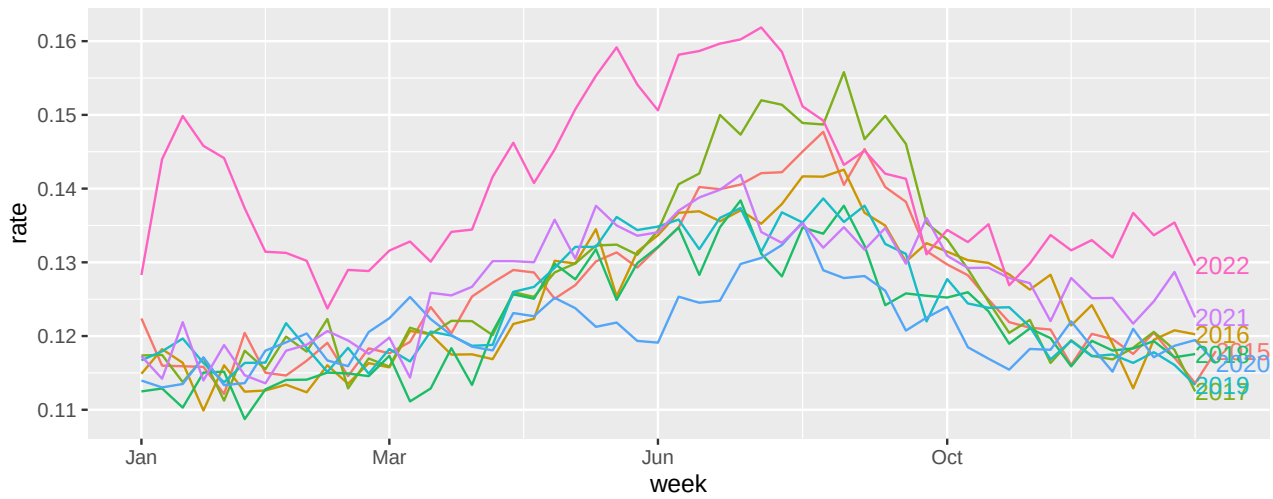

Figure 2

1. Using Figures 1–3, describe the weekly mortality rate for Australia. Carefully comment on the interesting features of all three plots.

   6 marks

2. For the STL decomposition shown in Figure 3, discuss the effect of the window sizes chosen for the trend and seasonal components. How would the results have changed with smaller or larger values chosen in each case?

   4 marks

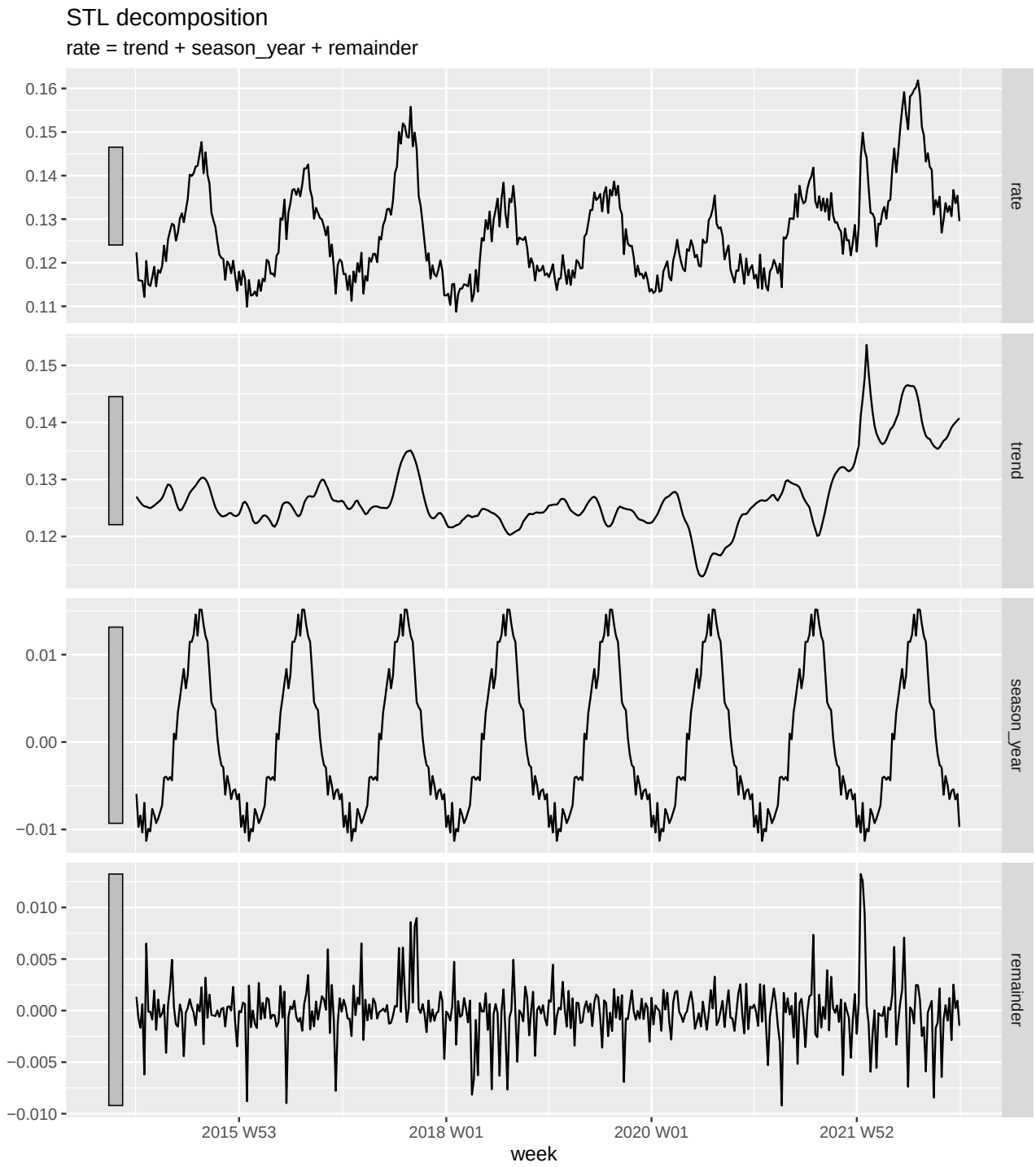
STL decomposition
rate = trend + season_year + remainder
0.16
0.15
0.14
0.13
0.12
0.11
rate
0.15
0.14
0.13
0.12
trend
0.01
0.00
−0.01
season_year
0.010
0.005
0.000
−0.005
−0.010
remainder
2015 W53
2018 W01
2020 W01
2021 W52
week

Figure 3

3. You have been asked to provide forecasts for the next year for Australian weekly mortality rates. Consider applying each of the methods and models below. Comment, in a few words each, on whether each one is appropriate for forecasting the data. No marks will be given for simply guessing whether a method or a model is appropriate without justifying your choice.

10 marks

Start your response by stating: **suitable** or **not suitable**.

(a) Seasonal naïve method using annual seasonality.
(b) Seasonal naïve method using weekly seasonality.
(c) An STL decomposition with an ETS to forecast the seasonally adjusted component, and seasonal naïve for the seasonal component.
(d) Holt's method with damped trend.
(e) ETS(A,N,A).
(f) ETS(M,A,M) with annual seasonality.
(g) ARIMA(0,1,52).
(h) ARIMA(2,0,2)$(0,1,0)_{52}$.
(i) Regression with Fourier terms for the annual seasonality.
(j) Dynamic regression with Fourier terms for the annual seasonality.

Total: 20 marks

SECTION C

1. Why can't you fit a seasonal ETS model to this data set?

3 marks

2. You decide to use the STL decomposition shown in Figure 3, with ETS used for the seasonally adjusted data.

```
my_model <- decomposition_model(
   STL(rate ~ season(window = "periodic")),
   ETS(season_adjust ~ season("N")),
   SNAIVE(season_year))

fit <- aus_mortality |>
   model(my_model)
report(fit)
```

```
Series: rate
Model: STL decomposition model
Combination: season_adjust + season_year

========================================

Series: season_adjust
Model: ETS(M,N,N)
  Smoothing parameters:
    alpha = 0.601

  Initial states:
  l[0]
 0.127

  sigma^2:  6e-04

  AIC  AICc   BIC
-2273 -2273 -2261

Series: season_year
Model: SNAIVE

sigma^2: 0
```

Write down the equations for the fitted model, including the STL equation, and the equations for the components.

6 marks

3. The residuals for the model are shown in Figure 4.

- What is causing the outlier in the residuals at the start of 2022?

2 marks

- Why are there no residuals for 2015?

2 marks

- Do you think the residuals would pass a white noise test?

2 marks

```
augment(fit) |>
   autoplot(.innov)
```

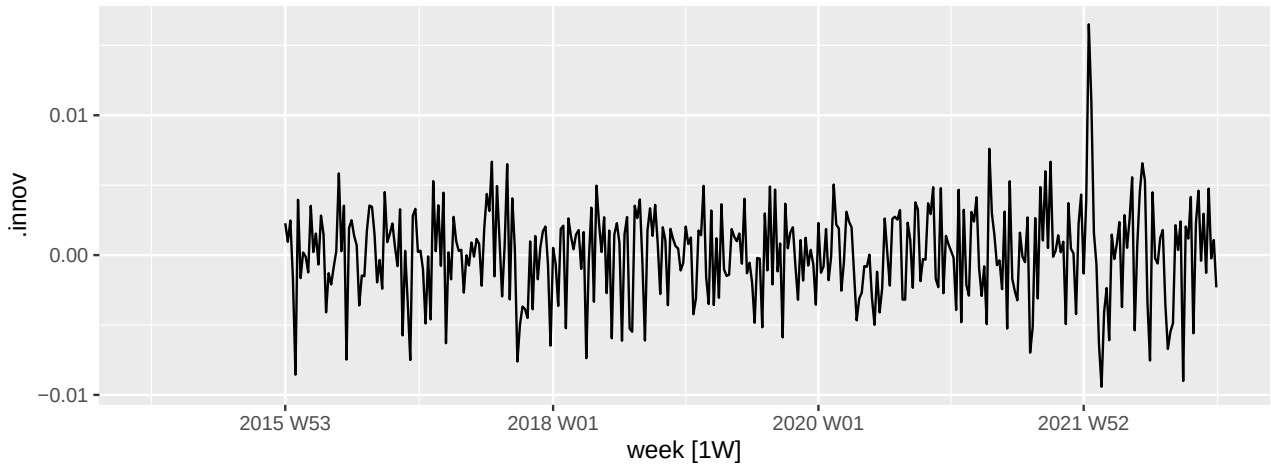

Figure 4

4. The forecasts for the model are shown in Figure 5.

```
fit |>
  forecast(h = "1 year") |>
  autoplot(aus_mortality)
```

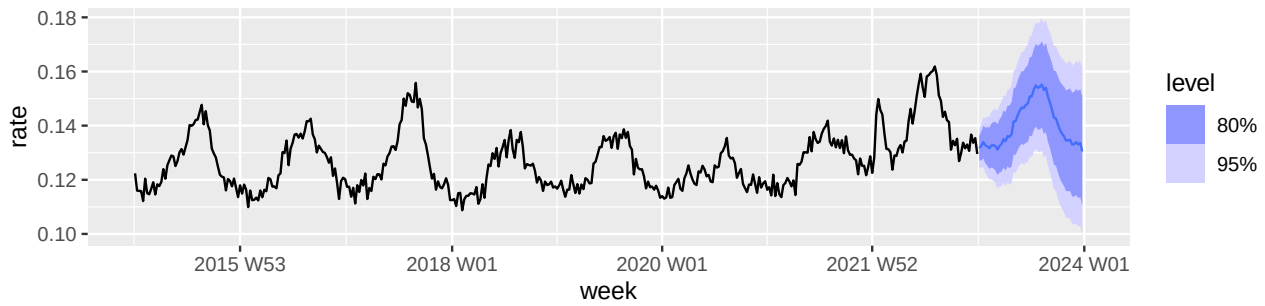

Figure 5

- What features of the data have not been captured by the model, and how does this affect the resulting forecasts?

3 marks

- Discuss how the large prediction intervals would affect how this model would be used to inform policy decisions around mortality risks?

2 marks

Total: 20 marks

SECTION D

An alternative model for the same data is a dynamic regression model, fitted as follows.

```r
fit <- aus_mortality |>
  model(ARIMA(rate ~ fourier(K = 2)))
report(fit)
```

```
Series: rate
Model: LM w/ ARIMA(2,1,1) errors

Coefficients:
         ar1     ar2     ma1  fourier(K = 2)C1_52  fourier(K = 2)S1_52
      0.5361  0.2474  -0.949               -5e-04              -0.0107
s.e.  0.0564  0.0526   0.027                1e-03               0.0010
      fourier(K = 2)C2_52  fourier(K = 2)S2_52
                  -0.0031                1e-04
s.e.               0.0007                7e-04

sigma^2 estimated as 1.173e-05:  log likelihood=1779
AIC=-3541   AICc=-3541   BIC=-3509
```

1. Write down the full model using backshift notation.

   4 marks

2. The number of pairs of Fourier terms was chosen by minimizing the AICc. Explain why this is a good way to choose a model being used for forecasting.

   2 marks

3. Comment on the model diagnostics shown in Figure 6 and the output below.

   5 marks

```r
fit |>
  gg_tsresiduals()
```

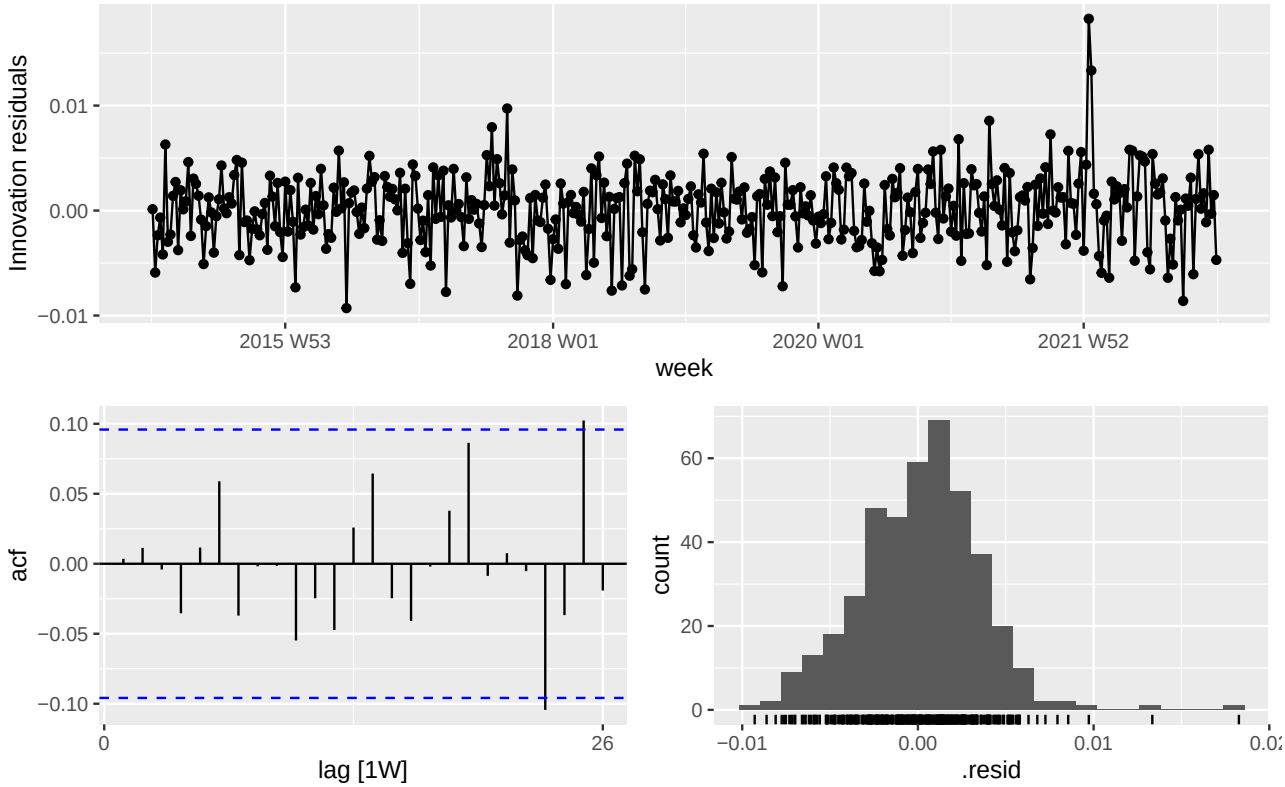

Figure 6:

```r
augment(fit) |>
  features(.innov, ljung_box, lag = 52, dof = 3)
```

```
# A tibble: 1 x 3
  .model                        lb_stat lb_pvalue
  <chr>                           <dbl>     <dbl>
1 ARIMA(rate ~ fourier(K = 2))     37.1     0.894
```

4. Figure 7 shows forecasts with prediction intervals for the next year.

```r
fit |>
  forecast(h = "1 year") |>
  autoplot(aus_mortality)
```

- The point forecasts for 2023 are lower than for 2022. What feature of the model causes this?

  2 marks

- If you wanted to allow for the COVID-19 pandemic, how could you modify the model?

  3 marks

- If you could obtain the actual weekly mortality rates for the first few months of 2023, describe in simple terms how you would measure the accuracy of the point forecasts and the accuracy of the prediction intervals from this model?

  4 marks

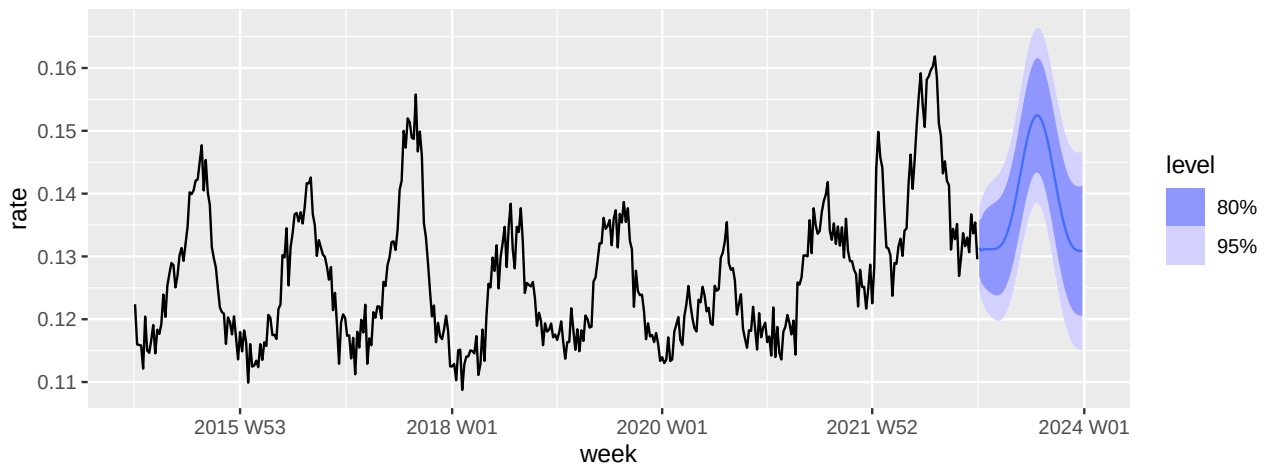

Figure 7:

Total: 20 marks

SECTION E

You decide to compare the two models discussed in Sections 3 and 4, along with a seasonal naïve benchmark. The following code uses a time series cross-validation with a 6 week test window.

```
my_model <- decomposition_model(
   STL(rate ~ season(window = "periodic")),
   ETS(season_adjust ~ season("N")),
   SNAIVE(season_year)
   )

stretch <- aus_mortality |>
   stretch_tsibble(.init = 105)

fit <- stretch |>
   model(
      stl_ets = my_model,
      dyn_regr = ARIMA(rate ~ fourier(K = 2)),
      snaive = SNAIVE(rate)
  )
```

```
fc1 <- fit |> forecast(h = "6 weeks")

fc1 |>
   accuracy(aus_mortality,
            measures = list(point_accuracy_measures, interval_accuracy_measures)
  )
```

```
# A tibble: 3 x 13
  .model   .type       ME    RMSE     MAE    MPE  MAPE  MASE RMSSE  ACF1 winkler
  <chr>    <chr>    <dbl>   <dbl>   <dbl>  <dbl> <dbl> <dbl> <dbl> <dbl>   <dbl>
1 dyn_regr Test  0.000236 0.00542 0.00394 0.0808  3.04 0.613 0.623 0.689  0.0337
2 snaive   Test  0.00261  0.00922 0.00685 1.73    5.18 1.07  1.06  0.798  0.0571
3 stl_ets  Test  0.000186 0.00547 0.00405 0.0849  3.14 0.631 0.628 0.638  0.0347
# i 2 more variables: pinball <dbl>, scaled_pinball <dbl>
```

1. The minimal training set used in the time series cross-validation included 105 weeks. Why couldn't this be made any smaller?

   2 marks

2. What do you conclude from the above output about the three models?

   2 marks

3. Even before the COVID-19 pandemic, each year showed a different pattern from the previous year, especially in terms of the size of the peak. This is due to seasonal illness such as flu. What aspects of the STL+ETS and dynamic regression models allow for these differences between years?

   4 marks

4. The accuracy statistics show that the mean error is smallest for the STL+ETS model, but the mean absolute error is smallest for the dynamic regression model. Why would we prefer to select a model using MAE rather than ME?

   3 marks

5. The MASE and RMSSE accuracy measures use scaled errors. Why is scaling errors unnecessary in this analysis?

2 marks

6. The MAPE accuracy measure uses percentage errors. While percentage errors are meaningful in this example, sometimes they are not. When would the MAPE give meaningless or unhelpful results?

2 marks

7. The Winkler score is defined as

$$W_{\alpha,t} = \begin{cases} (u_{\alpha,t} - \ell_{\alpha,t}) + \frac{2}{\alpha}(\ell_{\alpha,t} - y_t) & \text{if } y_t < \ell_{\alpha,t} \\ (u_{\alpha,t} - \ell_{\alpha,t}) & \text{if } \ell_{\alpha,t} \leq y_t \leq u_{\alpha,t} \\ (u_{\alpha,t} - \ell_{\alpha,t}) + \frac{2}{\alpha}(y_t - u_{\alpha,t}) & \text{if } y_t > u_{\alpha,t}. \end{cases}$$

where the $100(1-\alpha)\%$ prediction interval is given by $[\ell_{\alpha,t}, u_{\alpha,t}]$. The output above shows the Winkler score for a 95% prediction interval. Explain in words how the Winkler score works in terms of the width of the prediction interval and the penalties that apply when the observation is outside the prediction interval.

5 marks

Total: 20 marks